

Supplementary materials for the paper "Integrating Regularized Covariance Matrix Estimators" - the proof of Theorem 1, 2, 3, 4, 5, 6 and 7 in the paper.

Proof of Theorem 1. Define $R(\delta, \mathbf{D})$ as the squared Frobenius norm of $\boldsymbol{\Sigma}(\delta, \mathbf{D}) - \boldsymbol{\Sigma}_0$ in equation (2.2) in the paper. Then

$$\begin{aligned} R(\delta, \mathbf{D}) &= \left\| (1 - \delta) \mathbf{P} \mathbf{D} \mathbf{P}^T + \delta \mathbf{T} - \boldsymbol{\Sigma}_0 \right\|_F^2 \\ &= \left\| (1 - \delta) (\mathbf{P} \mathbf{D} \mathbf{P}^T - \boldsymbol{\Sigma}_0) + \delta (\mathbf{T} - \boldsymbol{\Sigma}_0) \right\|_F^2 \\ &= (1 - \delta)^2 \left\| \mathbf{P} \mathbf{D} \mathbf{P}^T - \boldsymbol{\Sigma}_0 \right\|_F^2 + 2\delta(1 - \delta) \text{tr}[(\mathbf{P} \mathbf{D} \mathbf{P}^T - \boldsymbol{\Sigma}_0)(\mathbf{T} - \boldsymbol{\Sigma}_0)] + \delta^2 \left\| \mathbf{T} - \boldsymbol{\Sigma}_0 \right\|_F^2. \end{aligned} \quad (\text{S.1})$$

First, we calculate the solution for $\mathbf{D} = \text{diag}(d_1, \dots, d_p)$ to the minimization problem in terms of δ . For each $j = 1, \dots, p$, we have for $\mathbf{P} = (\mathbf{p}_1, \dots, \mathbf{p}_p)$,

$$\begin{aligned} \frac{\partial R(\delta, \mathbf{D})}{\partial d_j} &= (1 - \delta)^2 \frac{\partial}{\partial d_j} (\text{tr}(\mathbf{P} \mathbf{D} \mathbf{P}^T \mathbf{P} \mathbf{D} \mathbf{P}^T) - 2\text{tr}(\mathbf{P} \mathbf{D} \mathbf{P}^T \boldsymbol{\Sigma}_0)) + 2\delta(1 - \delta) \frac{\partial}{\partial d_j} (\mathbf{P} \mathbf{D} \mathbf{P}^T (\mathbf{T} - \boldsymbol{\Sigma}_0)) \\ &= (1 - \delta)^2 \frac{\partial}{\partial d_j} \left(\sum_{i=1}^p d_i^2 - 2 \sum_{i=1}^p d_i \mathbf{p}_i^T \boldsymbol{\Sigma}_0 \mathbf{p}_i \right) + 2\delta(1 - \delta) \frac{\partial}{\partial d_j} \left(\sum_{i=1}^p d_i \mathbf{p}_i^T (\mathbf{T} - \boldsymbol{\Sigma}_0) \mathbf{p}_i \right) \\ &= (1 - \delta)^2 (2d_j - 2\mathbf{p}_j^T \boldsymbol{\Sigma}_0 \mathbf{p}_j) + 2\delta(1 - \delta) \mathbf{p}_j^T (\mathbf{T} - \boldsymbol{\Sigma}_0) \mathbf{p}_j \\ &= 2[(1 - \delta)^2 d_j - (1 - \delta)^2 \mathbf{p}_j^T \boldsymbol{\Sigma}_0 \mathbf{p}_j + \delta(1 - \delta) \mathbf{p}_j^T (\mathbf{T} - \boldsymbol{\Sigma}_0) \mathbf{p}_j]. \end{aligned}$$

Set $\frac{\partial R(\delta, \mathbf{D})}{\partial d_j} = 0$, then since we assumed $\delta \neq 1$,

$$\begin{aligned} d_j &= \frac{(1 - \delta)^2 \mathbf{p}_j^T \boldsymbol{\Sigma}_0 \mathbf{p}_j - \delta(1 - \delta) \mathbf{p}_j^T (\mathbf{T} - \boldsymbol{\Sigma}_0) \mathbf{p}_j}{(1 - \delta)^2} \\ &= \mathbf{p}_j^T \boldsymbol{\Sigma}_0 \mathbf{p}_j - \frac{\delta}{1 - \delta} (\mathbf{p}_j^T \mathbf{T} \mathbf{p}_j - \mathbf{p}_j^T \boldsymbol{\Sigma}_0 \mathbf{p}_j) \\ &= \frac{1}{1 - \delta} \mathbf{p}_j^T \boldsymbol{\Sigma}_0 \mathbf{p}_j - \frac{\delta}{1 - \delta} \mathbf{p}_j^T \mathbf{T} \mathbf{p}_j, \quad \text{so that} \\ \mathbf{D} &= \frac{1}{1 - \delta} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) - \frac{\delta}{1 - \delta} \text{diag}(\mathbf{P}^T \mathbf{T} \mathbf{P}). \end{aligned} \quad (\text{S.2})$$

This proves the first part of Theorem 1. For the second part of the proof, we substitute $\mathbf{D}$ from (S.2) into the original function $R(\delta, \mathbf{D})$ in (S.1). Hence

$$\begin{aligned} R(\delta, \mathbf{D}) &= R(\delta) = \left\| \mathbf{P} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) \mathbf{P}^T - \delta \mathbf{P} \text{diag}(\mathbf{P}^T \mathbf{T} \mathbf{P}) \mathbf{P}^T + \delta \mathbf{T} - \boldsymbol{\Sigma}_0 \right\|_F^2 \\ &= \left\| \mathbf{P} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) \mathbf{P}^T - \boldsymbol{\Sigma}_0 \right\|_F^2 + \delta^2 \left\| \hat{\boldsymbol{\Sigma}}_{\mathbf{T}} - \mathbf{T} \right\|_F^2 - 2\delta \text{tr}[(\mathbf{P} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) \mathbf{P}^T - \boldsymbol{\Sigma}_0)(\hat{\boldsymbol{\Sigma}}_{\mathbf{T}} - \mathbf{T})]. \\ \frac{\partial R(\delta)}{\partial \delta} &= 2\delta \left\| \hat{\boldsymbol{\Sigma}}_{\mathbf{T}} - \mathbf{T} \right\|_F^2 - 2\text{tr}[(\mathbf{P} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) \mathbf{P}^T - \boldsymbol{\Sigma}_0)(\hat{\boldsymbol{\Sigma}}_{\mathbf{T}} - \mathbf{T})] \\ &= 2[\delta \text{tr}(\hat{\boldsymbol{\Sigma}}_{\mathbf{T}} - \mathbf{T})^2 - \text{tr}(\boldsymbol{\Sigma}_0(\hat{\boldsymbol{\Sigma}}_{\mathbf{T}} - \mathbf{T}))], \end{aligned} \quad (\text{S.3})$$

where $\widehat{\Sigma}_{\mathbf{T}} = \mathbf{P} \text{diag}(\mathbf{P}^T \mathbf{T} \mathbf{P}) \mathbf{P}$, and we arrive at the equality in (S.3) since

$$\begin{aligned} \text{tr}(\mathbf{P} \text{diag}(\mathbf{P}^T \Sigma_0 \mathbf{P}) \mathbf{P}^T (\widehat{\Sigma}_{\mathbf{T}} - \mathbf{T})) &= \text{tr}(\mathbf{P} \text{diag}(\mathbf{P}^T \Sigma_0 \mathbf{P}) \mathbf{P}^T \mathbf{P} \text{diag}(\mathbf{P}^T \mathbf{T} \mathbf{P}) \mathbf{P}^T) - \text{tr}(\mathbf{P} \text{diag}(\mathbf{P}^T \Sigma_0 \mathbf{P}) \mathbf{P}^T \mathbf{T}) \\ &= \text{tr}(\text{diag}(\mathbf{P}^T \Sigma_0 \mathbf{P}) \text{diag}(\mathbf{P}^T \mathbf{T} \mathbf{P})) - \text{tr}(\text{diag}(\mathbf{P}^T \Sigma_0 \mathbf{P}) \mathbf{P}^T \mathbf{T} \mathbf{P}) \\ &= 0. \end{aligned} \quad (\text{S.4})$$

Setting $\frac{\partial R(\delta)}{\partial \delta} = 0$, we have the result as stated in Theorem 1, since $\text{tr}(\mathbf{T} - \widehat{\Sigma}_{\mathbf{T}})^2 \neq 0$ as $\mathbf{T}$ is not of the form $\mathbf{P} \mathbf{D} \mathbf{P}^T$ for some diagonal matrix $\mathbf{D}$. $\square$

To prove Theorem 2, we need to state a lemma first, which closely resembles Lemma 1 of Lam (2016).

Lemma 1. *Let Assumptions (A1) be satisfied. If the split location m is such that $\sum_{n \geq 1} p(n-m)^{-5} < \infty$, then we have*

$$\max_{1 \leq i \leq p} \left| \frac{\mathbf{q}_i^T \widetilde{\Sigma}_2 \mathbf{q}_i - \mathbf{q}_i^T \Sigma_0 \mathbf{q}_i}{\mathbf{q}_i^T \Sigma_0 \mathbf{q}_i} \right| \rightarrow 0$$

almost surely, where $\mathbf{q}_1, \dots, \mathbf{q}_p$ are unit vectors independent of the data $\mathbf{Y}_2$. The same holds true of the data is from a factor model, with Assumption (F1) satisfied together with $\sum_{n \geq 1} p(n-m)^{-5} < \infty$.

The proof of this lemma is exactly the same as that for Lemma 1 of Lam (2016), the only difference is the substitution of $\mathbf{p}_{1i}$ there by $\mathbf{q}_i$ here.

Proof of Theorem 2. With $\mathbf{T} - \widetilde{\Sigma}_{\mathbf{T}}$ being real symmetric, assume that $\mathbf{T} - \widetilde{\Sigma}_{\mathbf{T}} = \mathbf{Q} \mathbf{D}_{\mathbf{T}} \mathbf{Q}^T$, where $\mathbf{Q} = (\mathbf{q}_1, \dots, \mathbf{q}_p)$ is orthogonal and $\mathbf{D}_{\mathbf{T}} = \text{diag}(d_1, \dots, d_p)$ is diagonal. The $\mathbf{q}_i$'s are independent of $\mathbf{Y}_2$ since $\mathbf{T}$ is constructed from the data $\mathbf{Y}_1$. Then with Assumptions (A1) to (A2) assuming the data is not from a factor model,

$$\begin{aligned} |\widehat{\delta} - \delta| &= \frac{|\text{tr}[(\mathbf{T} - \widetilde{\Sigma}_{\mathbf{T}})(\widetilde{\Sigma}_2 - \Sigma_0)]|}{\text{tr}(\mathbf{T} - \widetilde{\Sigma}_{\mathbf{T}})^2} = \frac{|\sum_{i=1}^p d_i (\mathbf{q}_i^T \widetilde{\Sigma}_2 \mathbf{q}_i - \mathbf{q}_i^T \Sigma_0 \mathbf{q}_i)|}{\text{tr}(\mathbf{T} - \widetilde{\Sigma}_{\mathbf{T}})^2} \\ &\leq \frac{\left(p^{-1} \sum_{i=1}^p d_i^2\right)^{1/2} \left(p^{-1} \sum_{i=1}^p (\mathbf{q}_i^T \widetilde{\Sigma}_2 \mathbf{q}_i - \mathbf{q}_i^T \Sigma_0 \mathbf{q}_i)^2\right)^{1/2}}{p^{-1} \text{tr}(\mathbf{T} - \widetilde{\Sigma}_{\mathbf{T}})^2} \\ &\leq \frac{\max_{1 \leq i \leq p} \left| \frac{\mathbf{q}_i^T \widetilde{\Sigma}_2 \mathbf{q}_i - \mathbf{q}_i^T \Sigma_0 \mathbf{q}_i}{\mathbf{q}_i^T \Sigma_0 \mathbf{q}_i} \right| \left(p^{-1} \sum_{i=1}^p (\mathbf{q}_i^T \Sigma_0 \mathbf{q}_i)^2\right)^{1/2}}{(p^{-1} \text{tr}(\mathbf{T} - \widetilde{\Sigma}_{\mathbf{T}})^2)^{1/2}} \rightarrow 0 \end{aligned} \quad (\text{S.5})$$

in probability/almost surely by Lemma 1, the fact that $(p^{-1} \sum_{i=1}^p (\mathbf{q}_i^T \Sigma_0 \mathbf{q}_i)^2)^{1/2} \leq \lambda_{\max}(\Sigma_0) < \infty$ by Assumption (A2), and finally by the assumption that $p^{-1} \text{tr}(\mathbf{T} - \widetilde{\Sigma}_{\mathbf{T}})^2 \rightarrow 0$ in probability/almost surely. This proves $\widehat{\delta} - \delta \rightarrow 0$ in probability/almost surely.

Now consider

$$\begin{aligned}\hat{\delta} &= \frac{\text{tr}[(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})\Sigma_0]}{\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2} = \frac{\text{tr}[(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})(\Sigma_0 - \mathbf{T})] + \text{tr}[(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})\mathbf{T}]}{\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2} \\ &= 1 + \frac{\text{tr}[(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})(\Sigma_0 - \mathbf{T})]}{\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2},\end{aligned}$$

where the last line follows since $\text{tr}[(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})\tilde{\Sigma}_{\mathbf{T}}] = 0$. But with the assumption $\|\mathbf{T} - \Sigma_0\| \rightarrow 0$ in probability/almost surely, then

$$\left| \frac{\text{tr}[(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})(\Sigma_0 - \mathbf{T})]}{\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2} \right| \leq \frac{(p^{-1}\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2)^{1/2} (p^{-1}\text{tr}(\mathbf{T} - \Sigma_0)^2)^{1/2}}{p^{-1}\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2} \leq \frac{\|\mathbf{T} - \Sigma_0\|}{(p^{-1}\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2)^{1/2}} \rightarrow 0$$

in probability/almost surely by the assumption that $p^{-1}\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2 \rightarrow 0$ in probability/almost surely. This proves that $\hat{\delta} \rightarrow 1$ in probability/almost surely. With this, then

$$\begin{aligned}\|\Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2) - \Sigma_0\| &= \|\mathbf{P}_1 \text{diag}(\mathbf{P}_1^T \tilde{\Sigma}_2 \mathbf{P}_1) \mathbf{P}_1^T + \hat{\delta}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}}) - \Sigma_0\| \\ &\leq \|\mathbf{P}_1 \text{diag}(\mathbf{P}_1^T \tilde{\Sigma}_2 \mathbf{P}_1) \mathbf{P}_1^T - \tilde{\Sigma}_{\mathbf{T}}\| + \|\mathbf{T} - \Sigma_0\| + |\hat{\delta} - 1| \cdot \|\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}}\| \\ &= \max_{1 \leq i \leq p} |\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \mathbf{T} \mathbf{p}_{1i}| + \|\mathbf{T} - \Sigma_0\| + |\hat{\delta} - 1| \cdot \|\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}}\| \\ &\leq \max_{1 \leq i \leq p} \left| \frac{\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_0 \mathbf{p}_{1i}}{\mathbf{p}_{1i}^T \Sigma_0 \mathbf{p}_{1i}} \right| \cdot \|\Sigma_0\| + \max_{1 \leq i \leq p} |\mathbf{p}_{1i}^T \mathbf{T} \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_0 \mathbf{p}_{1i}| \\ &\quad + \|\mathbf{T} - \Sigma_0\| + 2|\hat{\delta} - 1| \cdot \|\mathbf{T}\| \\ &\leq \max_{1 \leq i \leq p} \left| \frac{\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_0 \mathbf{p}_{1i}}{\mathbf{p}_{1i}^T \Sigma_0 \mathbf{p}_{1i}} \right| \cdot \|\Sigma_0\| + 2\|\mathbf{T} - \Sigma_0\| + 2|\hat{\delta} - 1| \cdot \|\mathbf{T}\| \rightarrow 0\end{aligned}$$

in probability/almost surely, where the last line follows from Lemma 1 of Lam (2016), Assumption (A2) that $\|\Sigma_0\|$ is finite, and the assumption that $\|\mathbf{T}\|$ is finite in probability/almost surely.

For data from a factor model, almost every details of the proofs are exactly the same as before, except that we now have $p^{-2}\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2 \rightarrow 0$ while $p^{-1}\|\Sigma_0\| = O(1)$ and $p^{-1}\|\mathbf{T}\|$ are finite in probability/almost surely. This completes the proof of the theorem. $\square$

To prove Theorem 3, we need to state and prove another lemma first.

Lemma 2. *Let Assumptions (A1) to (A4) be satisfied, and $\Sigma_0 \neq \sigma^2 \mathbf{I}_p$. Assume the split location $m = m(n)$ satisfies the constraints $m/n \rightarrow 1$, $n - m \rightarrow \infty$ and $\sum_{n \geq 1} p(n - m)^{-5} < \infty$ while $p = p(n)$ satisfies $p/n \rightarrow c > 0$. Then we have $\|\mathbf{P}_1 - \mathbf{P}\| \rightarrow 0$ almost surely. In particular, assuming $\|\mathbf{T}\|$ finite in*

probability/almost surely, we have in probability/almost surely,

$$\frac{1}{p} \sum_{i=1}^p |\mathbf{p}_{1i}^T \mathbf{T} \mathbf{p}_{1i} - \mathbf{p}_i^T \mathbf{T} \mathbf{p}_i| \rightarrow 0, \quad \frac{1}{p} \sum_{i=1}^p |\mathbf{p}_{1i}^T \mathbf{\Sigma}_0 \mathbf{p}_{1i} - \mathbf{p}_i^T \mathbf{\Sigma}_0 \mathbf{p}_i| \rightarrow 0.$$

Proof of Lemma 2. The setting is the same as that in Lemma S.4 of Lam (2016). As such, for λ_i and λ_{1i} being the eigenvalues of $\tilde{\mathbf{\Sigma}} = n^{-1} \mathbf{Y} \mathbf{Y}^T$ and $\tilde{\mathbf{\Sigma}}_1 = m^{-1} \mathbf{Y}_1 \mathbf{Y}_1^T$ respectively (with corresponding eigenvectors $\mathbf{p}_i$ and $\mathbf{p}_{1i}$), we have for any continuous function $g(\cdot)$ over the positive real line,

$$p^{-1} \sum_{i=1}^p g(\mathbf{p}_i^T \mathbf{\Sigma}_0 \mathbf{p}_i) \mathbf{1}_{\{\lambda_i \leq x\}} \xrightarrow{a.s.} \int_{\infty}^x g(\delta(\lambda)) dF(\lambda), \quad (\text{S.6})$$

where $F(\cdot)$ is such that

$$F_p(\lambda) = p^{-1} \sum_{i=1}^p \mathbf{1}_{\{\lambda_i \leq \lambda\}} \xrightarrow{a.s.} F(\lambda).$$

Equation (S.6) is the same as equation (S.3) in Lam (2016). Please refer to equation (2.7) of Lam (2016) for the definition of $\delta(\cdot)$. Similarly, we have

$$p^{-1} \sum_{i=1}^p g(\mathbf{p}_{1i}^T \mathbf{\Sigma}_0 \mathbf{p}_{1i}) \mathbf{1}_{\{\lambda_{1i} \leq x\}} \xrightarrow{a.s.} \int_{\infty}^x g(\delta_1(\lambda)) dF_1(\lambda), \quad (\text{S.7})$$

where $\delta_1(\cdot)$ is as in (2.9) of Lam (2016), and $F_1(\cdot)$ is such that

$$F_{1p}(\lambda) = p^{-1} \sum_{i=1}^p \mathbf{1}_{\{\lambda_{1i} \leq \lambda\}} \xrightarrow{a.s.} F_1(\lambda).$$

But since $m/n \rightarrow 1$, we have $p/m, p/n$ both go to the same limit $c > 0$. Theorem 4.1 of Bai and Silverstein (2010) tells us then both F_p and F_{1p} converge to the same limit almost surely under Assumptions (A1) to (A4). Hence $F = F_1$ almost surely, implying $\delta_1(\cdot) = \delta(\cdot)$ almost surely. Setting $g \equiv 1$, we see that λ_{1i} and λ_i are almost surely the same.

Then using Theorem 3 of Ledoit and P  ch   (2011), we can arrive at that both $\mathbf{p}_{1i}^T \mathbf{v}_j$ and $\mathbf{p}_i^T \mathbf{v}_j$ for $i, j = 1, \dots, p$ are almost surely the same as a function $\varphi(\lambda_i, \tau_j)$ uniformly for each i and j (which depends on the same constant c where $p/m, p/n \rightarrow c > 0$ since $m/n \rightarrow 1$), where $\mathbf{v}_j$ is the eigenvector of $\mathbf{\Sigma}_0$ corresponding to the j -th largest eigenvalue τ_j of $\mathbf{\Sigma}_0$. Hence

$$\max_{1 \leq i, j \leq p} |(\mathbf{p}_{1i} - \mathbf{p}_i)^T \mathbf{v}_j| \xrightarrow{a.s.} 0.$$

It means that we must have $\max_{1 \leq i \leq p} \|\mathbf{p}_{1i} - \mathbf{p}_i\| \xrightarrow{a.s.} 0$. Hence $\|\mathbf{P}_1 - \mathbf{P}\| \xrightarrow{a.s.} 0$ follows. With this, then

$$\frac{1}{p} \sum_{i=1}^p |\mathbf{p}_{1i}^\top \mathbf{T} \mathbf{p}_{1i} - \mathbf{p}_i^\top \mathbf{T} \mathbf{p}_i| = \frac{1}{p} \sum_{i=1}^p |(\mathbf{p}_{1i} - \mathbf{p}_i)^\top \mathbf{T} (\mathbf{p}_{1i} + \mathbf{p}_i)| \leq 2 \|\mathbf{T}\| \max_{1 \leq i \leq p} \|\mathbf{p}_{1i} - \mathbf{p}_i\| \rightarrow 0$$

in probability/almost surely, since $\|\mathbf{T}\|$ is finite in probability/almost surely. Replacing $\mathbf{T}$ by $\mathbf{\Sigma}_0$ proves the almost sure convergence of the other by Assumption (A2) that $\|\mathbf{\Sigma}_0\| = O(1)$. $\square$

Proof of Theorem 3. Recall the notations $\hat{\mathbf{\Sigma}}_{\mathbf{T}} = \mathbf{P} \text{diag}(\mathbf{P}^\top \mathbf{T} \mathbf{P}) \mathbf{P}^\top$ and $\tilde{\mathbf{\Sigma}}_{\mathbf{T}} = \mathbf{P}_1 \text{diag}(\mathbf{P}_1^\top \mathbf{T} \mathbf{P}_1) \mathbf{P}_1^\top$. Consider

$$\begin{aligned} \frac{1}{p} \left\| \mathbf{\Sigma}_{\text{Ideal}} - \mathbf{\Sigma}_0 \right\|_F^2 &= \frac{1}{p} \left\| \mathbf{P} \text{diag}(\mathbf{P}^\top \mathbf{\Sigma}_0 \mathbf{P}) \mathbf{P}^\top - \mathbf{\Sigma}_0 + \frac{\text{tr}[(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}}) \mathbf{\Sigma}_0]}{\text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2} (\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}}) \right\|_F^2 \\ &= \frac{1}{p} \left\| \mathbf{P} \text{diag}(\mathbf{P}^\top \mathbf{\Sigma}_0 \mathbf{P}) \mathbf{P}^\top - \mathbf{\Sigma}_0 \right\|_F^2 + \frac{1}{p} \frac{\text{tr}^2[(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}}) \mathbf{\Sigma}_0]}{\text{tr}^2(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2} \text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2 \\ &\quad + \frac{2}{p} \text{tr}[(\mathbf{P} \text{diag}(\mathbf{P}^\top \mathbf{\Sigma}_0 \mathbf{P}) \mathbf{P}^\top - \mathbf{\Sigma}_0)(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})] \frac{\text{tr}[(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}}) \mathbf{\Sigma}_0]}{\text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2} \\ &= \frac{1}{p} \left\| \text{diag}(\mathbf{P}^\top \mathbf{\Sigma}_0 \mathbf{P}) - \mathbf{P}^\top \mathbf{\Sigma}_0 \mathbf{P} \right\|_F^2 - \frac{1}{p} \frac{\text{tr}^2[(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}}) \mathbf{\Sigma}_0]}{\text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2}. \end{aligned}$$

Similarly, we have

$$\begin{aligned} \frac{1}{p} \left\| \mathbf{\Sigma}(\mathbf{P}_1, \mathbf{T}, \tilde{\mathbf{\Sigma}}_2) - \mathbf{\Sigma}_0 \right\|_F^2 &= \frac{1}{p} \left\| \text{diag}(\mathbf{P}_1^\top \tilde{\mathbf{\Sigma}}_2 \mathbf{P}) - \mathbf{P}_1^\top \mathbf{\Sigma}_0 \mathbf{P}_1 \right\|_F^2 + \frac{1}{p} \frac{\text{tr}^2[(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}}) \tilde{\mathbf{\Sigma}}_2]}{\text{tr}(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})^2} \\ &\quad - \frac{2}{p} \frac{\text{tr}[(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}}) \tilde{\mathbf{\Sigma}}_2]}{\text{tr}(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})^2} \text{tr}[(\mathbf{T} - \tilde{\mathbf{\Sigma}}_2) \mathbf{\Sigma}_0] \\ &= \frac{1}{p} \sum_{i=1}^p (\mathbf{p}_{1i}^\top \tilde{\mathbf{\Sigma}}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^\top \mathbf{\Sigma}_0 \mathbf{p}_{1i})^2 + \frac{1}{p} \left\| \text{diag}(\mathbf{P}_1^\top \mathbf{\Sigma}_0 \mathbf{P}) - \mathbf{P}_1^\top \mathbf{\Sigma}_0 \mathbf{P}_1 \right\|_F^2 \\ &\quad - \frac{1}{p} \frac{\text{tr}^2[(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}}) \mathbf{\Sigma}_0]}{\text{tr}(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})^2} + \frac{1}{p} \frac{\text{tr}^2[(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})(\tilde{\mathbf{\Sigma}}_2 - \mathbf{\Sigma}_0)]}{\text{tr}(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})^2}. \end{aligned}$$

With these, we can expand the efficiency loss as

$$\begin{aligned}
EL(\Sigma_0, \Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2)) &= 1 - \left(\frac{p^{-1} \left\| \Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2) - \Sigma_0 \right\|_F^2}{p^{-1} \left\| \Sigma_{\text{Ideal}} - \Sigma_0 \right\|_F^2} \right)^{-1} \\
&= 1 - \left(\frac{R_1 + R_2 + R_3 + R_4}{p^{-1} \left\| \Sigma_{\text{Ideal}} - \Sigma_0 \right\|_F^2} \right)^{-1}, \quad \text{where} \\
R_1 &= \frac{1}{p} \left\| \text{diag}(\mathbf{P}_1^T \Sigma_0 \mathbf{P}_1) - \mathbf{P}_1^T \Sigma_0 \mathbf{P}_1 \right\|_F^2 - \frac{1}{p} \frac{\text{tr}^2[(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}}) \Sigma_0]}{\text{tr}(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})^2}, \\
R_2 &= \frac{1}{p} \sum_{i=1}^p (\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_0 \mathbf{p}_{1i})^2, \\
R_3 &= \frac{1}{p} \frac{\text{tr}^2[(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}}) \Sigma_0]}{\text{tr}(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})^2} - \frac{1}{p} \frac{\text{tr}^2[(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}}) \Sigma_0]}{\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2}, \\
R_4 &= \frac{1}{p} \frac{\text{tr}^2[(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})(\tilde{\Sigma}_2 - \Sigma_0)]}{\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2}.
\end{aligned}$$

Firstly, we have $\frac{R_1}{p^{-1} \left\| \Sigma_{\text{Ideal}} - \Sigma_0 \right\|_F^2} \rightarrow 1$ almost surely by Lemma S.4 of Lam (2016). If we can prove that $R_2, R_3, R_4 \rightarrow 0$ in probability/almost surely, then since $p^{-1} \left\| \Sigma_{\text{Ideal}} - \Sigma_0 \right\|_F^2 \not\rightarrow 0$ as $\Sigma_0 \neq \sigma^2 \mathbf{I}_p$ (see the proof of Lemma S.4 in Lam (2016) for more details), the proof will be completed.

To this end, apply Lemma 1 with $\mathbf{q}_i = \mathbf{p}_{1i}$,

$$\begin{aligned}
R_2 &\leq \left(\max_{1 \leq i \leq p} \left| \frac{\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_0 \mathbf{p}_{1i}}{\mathbf{p}_{1i}^T \Sigma_0 \mathbf{p}_{1i}} \right| \right)^2 \cdot \max_{1 \leq i \leq p} (\mathbf{p}_{1i}^T \Sigma_0 \mathbf{p}_{1i})^2 \\
&\leq \left(\max_{1 \leq i \leq p} \left| \frac{\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_0 \mathbf{p}_{1i}}{\mathbf{p}_{1i}^T \Sigma_0 \mathbf{p}_{1i}} \right| \right)^2 \cdot \left\| \Sigma_0 \right\|^2 \rightarrow 0
\end{aligned} \tag{S.8}$$

almost surely as $\left\| \Sigma_0 \right\| = O(1)$ by Assumption (A2). For the term R_3 , consider

$$\begin{aligned}
R_3 &= R_{3,1} + R_{3,2}, \quad \text{where} \\
R_{3,1} &= \frac{1}{p} (\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2 - \text{tr}(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})^2) \cdot \frac{\text{tr}^2(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}}) \Sigma_0}{\text{tr}(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})^2 \text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2}, \\
R_{3,2} &= \frac{1}{p} \frac{\text{tr}^2(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}}) \Sigma_0 - \text{tr}^2(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}}) \Sigma_0}{\text{tr}(\mathbf{T} - \hat{\Sigma}_{\mathbf{T}})^2}.
\end{aligned}$$

To bound $R_{3,1}$, consider

$$\begin{aligned}
|R_{3,1}| &= \frac{1}{p} |\text{tr}(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})^2 - \text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2| \cdot \frac{\text{tr}^2[(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})\mathbf{\Sigma}_0]}{\text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2 \text{tr}(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})^2} \\
&\leq \frac{1}{p} |\text{tr}(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})^2 - \text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2| \cdot \frac{p^{-1} \text{tr}(\mathbf{\Sigma}_0^2)}{p^{-1} \text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2} \\
&= \left| \frac{1}{p} (\text{tr}(\mathbf{T}^2) - \sum_{i=1}^p (\mathbf{p}_i^{\text{T}} \mathbf{T} \mathbf{p}_i)^2) - \frac{1}{p} (\text{tr}(\mathbf{T}^2) - \sum_{i=1}^p (\mathbf{p}_{1i}^{\text{T}} \mathbf{T} \mathbf{p}_{1i})^2) \right| \cdot \frac{p^{-1} \text{tr}(\mathbf{\Sigma}_0^2)}{p^{-1} \text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2} \\
&\leq \max_{1 \leq i \leq p} |\mathbf{p}_{1i}^{\text{T}} \mathbf{T} \mathbf{p}_{1i} + \mathbf{p}_i^{\text{T}} \mathbf{T} \mathbf{p}_i| \cdot \frac{1}{p} \sum_{i=1}^p |\mathbf{p}_{1i}^{\text{T}} \mathbf{T} \mathbf{p}_{1i} - \mathbf{p}_i^{\text{T}} \mathbf{T} \mathbf{p}_i| \cdot \frac{\|\mathbf{\Sigma}_0\|^2}{p^{-1} \text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2} \\
&\leq 2\|\mathbf{T}\| \cdot \frac{1}{p} \sum_{i=1}^p |\mathbf{p}_{1i}^{\text{T}} \mathbf{T} \mathbf{p}_{1i} - \mathbf{p}_i^{\text{T}} \mathbf{T} \mathbf{p}_i| \cdot \frac{\|\mathbf{\Sigma}_0\|^2}{p^{-1} \text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2} \rightarrow 0
\end{aligned}$$

in probability/almost surely by the assumption $p^{-1} \text{tr}(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})^2 \not\rightarrow 0$ in probability/almost surely with $\|\mathbf{T}\|$ being finite, that $\|\mathbf{\Sigma}_0\| = O(1)$ by Assumption (A2), and finally by the result of Lemma 2.

Also, we have

$$\begin{aligned}
\frac{1}{p} |\text{tr}[(\hat{\mathbf{\Sigma}}_{\mathbf{T}} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})\mathbf{\Sigma}_0]| &= \frac{1}{p} |\text{tr}(\text{diag}(\mathbf{P}^{\text{T}} \mathbf{T} \mathbf{P}) \mathbf{P}^{\text{T}} \mathbf{\Sigma}_0 \mathbf{P}) - \text{tr}(\text{diag}(\mathbf{P}_1^{\text{T}} \mathbf{T} \mathbf{P}_1) \mathbf{P}_1^{\text{T}} \mathbf{\Sigma}_0 \mathbf{P}_1)| \\
&\leq \frac{1}{p} \sum_{i=1}^p |\mathbf{p}_i^{\text{T}} \mathbf{T} \mathbf{p}_i| \cdot |\mathbf{p}_i^{\text{T}} \mathbf{\Sigma}_0 \mathbf{p}_i - \mathbf{p}_{1i}^{\text{T}} \mathbf{\Sigma}_0 \mathbf{p}_{1i}| + \frac{1}{p} \sum_{i=1}^p |\mathbf{p}_{1i}^{\text{T}} \mathbf{\Sigma}_0 \mathbf{p}_{1i}| \cdot |\mathbf{p}_i^{\text{T}} \mathbf{T} \mathbf{p}_i - \mathbf{p}_{1i}^{\text{T}} \mathbf{T} \mathbf{p}_{1i}| \\
&\leq \|\mathbf{T}\| \cdot \frac{1}{p} \sum_{i=1}^p |\mathbf{p}_i^{\text{T}} \mathbf{\Sigma}_0 \mathbf{p}_i - \mathbf{p}_{1i}^{\text{T}} \mathbf{\Sigma}_0 \mathbf{p}_{1i}| + \|\mathbf{\Sigma}_0\| \cdot \frac{1}{p} \sum_{i=1}^p |\mathbf{p}_i^{\text{T}} \mathbf{T} \mathbf{p}_i - \mathbf{p}_{1i}^{\text{T}} \mathbf{T} \mathbf{p}_{1i}| \rightarrow 0 \quad (\text{S.9})
\end{aligned}$$

in probability/almost surely by Lemma 2. We can then bound $R_{3,2}$ by noting that

$$\begin{aligned}
|R_{3,2}| &= \frac{1}{p} \frac{|\text{tr}^2[(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})\mathbf{\Sigma}_0] - \text{tr}^2[(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})\mathbf{\Sigma}_0]|}{\text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2} \\
&= \left| \frac{1}{p} \text{tr}[(\mathbf{T} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})\mathbf{\Sigma}_0] - \frac{1}{p} \text{tr}[(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})\mathbf{\Sigma}_0] \right| \cdot \frac{|2\text{tr}[(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})\mathbf{\Sigma}_0] + \text{tr}[(\hat{\mathbf{\Sigma}}_{\mathbf{T}} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})\mathbf{\Sigma}_0]|}{\text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2} \\
&\leq \frac{1}{p} |\text{tr}[(\hat{\mathbf{\Sigma}}_{\mathbf{T}} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})\mathbf{\Sigma}_0]| \cdot \left(\frac{2\|\mathbf{\Sigma}_0\|}{(p^{-1} \text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2)^{1/2}} + \frac{|p^{-1} \text{tr}[(\hat{\mathbf{\Sigma}}_{\mathbf{T}} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})\mathbf{\Sigma}_0]|}{p^{-1} \text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2} \right) \rightarrow 0
\end{aligned}$$

in probability/almost surely by what we have just proved that $p^{-1} \text{tr}[(\hat{\mathbf{\Sigma}}_{\mathbf{T}} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}})\mathbf{\Sigma}_0] \rightarrow 0$ in probability/almost surely, that $\|\mathbf{\Sigma}_0\| = O(1)$ by Assumption (A2), and the assumption that $p^{-1} \text{tr}(\mathbf{T} - \hat{\mathbf{\Sigma}}_{\mathbf{T}})^2 \not\rightarrow 0$ in probability/almost surely.

Combining, hence we have $R_3 = R_{3,1} + R_{3,2} \rightarrow 0$ in probability/almost surely.

Finally, similar to the proof of Theorem 2, assuming $\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}} = \mathbf{Q}\mathbf{D}_{\mathbf{T}}\mathbf{Q}^T$, we have

$$\begin{aligned} R_4 &= \frac{1}{p} \frac{[\sum_{i=1}^p d_i (\mathbf{q}_i^T \tilde{\Sigma}_2 \mathbf{q}_i - \mathbf{q}_i^T \Sigma_0 \mathbf{q}_i)]^2}{\text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2} \\ &\leq \frac{p^{-1} \sum_{i=1}^p (\mathbf{q}_i^T \tilde{\Sigma}_2 \mathbf{q}_i - \mathbf{q}_i^T \Sigma_0 \mathbf{q}_i)^2 \cdot p^{-1} \sum_{i=1}^p d_i^2}{p^{-1} \text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2} \\ &\leq \max_{1 \leq i \leq p} \left| \frac{\mathbf{q}_i^T \tilde{\Sigma}_2 \mathbf{q}_i - \mathbf{q}_i^T \Sigma_0 \mathbf{q}_i}{\mathbf{q}_i^T \Sigma_0 \mathbf{q}_i} \right| \cdot p^{-1} \sum_{i=1}^p (\mathbf{q}_i^T \Sigma_0 \mathbf{q}_i)^2 \rightarrow 0 \end{aligned}$$

in probability/almost surely by Lemma 1, the fact that $p^{-1} \text{tr}(\mathbf{T} - \tilde{\Sigma}_{\mathbf{T}})^2 \rightarrow 0$ in probability/almost surely by assumption, and $p^{-1} \sum_{i=1}^p (\mathbf{q}_i^T \Sigma_0 \mathbf{q}_i)^2 \leq \|\Sigma_0\|^2 < \infty$ by Assumption (A2). This completes the proof of the Theorem. $\square$

Proof of Theorem 4. Similar to the proof of Theorem 1, define $R(\delta_1, \delta_2, \mathbf{D})$ as the squared Frobenius norm of $\Sigma(\delta_1, \delta_2, \mathbf{D}) - \Sigma_0$ in equation (3.1) in the paper. Then

$$\begin{aligned} R(\delta_1, \delta_2, \mathbf{D}) &= (1 - \delta_1 - \delta_2)^2 \left\| \mathbf{P}\mathbf{D}\mathbf{P}^T - \Sigma_0 \right\|_F^2 + \delta_1^2 \left\| \mathbf{T}_1 - \Sigma_0 \right\|_F^2 + \delta_2^2 \left\| \mathbf{T}_2 - \Sigma_0 \right\|_F^2 \\ &\quad + 2\delta_1(1 - \delta_1 - \delta_2) \text{tr}[(\mathbf{P}\mathbf{D}\mathbf{P}^T - \Sigma_0)(\mathbf{T}_1 - \Sigma_0)] + 2\delta_1\delta_2 \text{tr}[(\mathbf{T}_1 - \Sigma_0)(\mathbf{T}_2 - \Sigma_0)] \\ &\quad + 2\delta_2(1 - \delta_1 - \delta_2) \text{tr}[(\mathbf{P}\mathbf{D}\mathbf{P}^T - \Sigma_0)(\mathbf{T}_2 - \Sigma_0)]. \end{aligned} \tag{S.10}$$

To minimize $R(\delta_1, \delta_2, \mathbf{D})$, we first calculate the solution for $\mathbf{D} = \text{diag}(d_1, \dots, d_p)$ in terms of δ_1 and δ_2 . For each $j = 1, \dots, p$, we have

$$\begin{aligned} \frac{\partial R(\delta_1, \delta_2, \mathbf{D})}{\partial d_j} &= (1 - \delta_1 - \delta_2)^2 \frac{\partial}{\partial d_j} (\text{tr}(\mathbf{P}\mathbf{D}\mathbf{P}^T \mathbf{P}\mathbf{D}\mathbf{P}^T) - 2\text{tr}(\mathbf{P}\mathbf{D}\mathbf{P}^T \Sigma_0)) \\ &\quad + 2\delta_1(1 - \delta_1 - \delta_2) \frac{\partial}{\partial d_j} \text{tr}(\mathbf{P}\mathbf{D}\mathbf{P}^T (\mathbf{T}_1 - \Sigma_0)) + 2\delta_2(1 - \delta_1 - \delta_2) \frac{\partial}{\partial d_j} \text{tr}(\mathbf{P}\mathbf{D}\mathbf{P}^T (\mathbf{T}_2 - \Sigma_0)) \\ &= (1 - \delta_1 - \delta_2)^2 \frac{\partial}{\partial d_j} \left(\sum_{i=1}^p d_i^2 - 2 \sum_{i=1}^p d_i \mathbf{p}_i^T \Sigma_0 \mathbf{p}_i \right) + 2\delta_1(1 - \delta_1 - \delta_2) \frac{\partial}{\partial d_j} \left(\sum_{i=1}^p d_i \mathbf{p}_i^T (\mathbf{T}_1 - \Sigma_0) \mathbf{p}_i \right) \\ &\quad + 2\delta_2(1 - \delta_1 - \delta_2) \frac{\partial}{\partial d_j} \left(\sum_{i=1}^p d_i \mathbf{p}_i^T (\mathbf{T}_2 - \Sigma_0) \mathbf{p}_i \right) \\ &= 2(1 - \delta_1 - \delta_2) [(1 - \delta_1 - \delta_2)(d_j - \mathbf{p}_j^T \Sigma_0 \mathbf{p}_j) + \delta_1 \mathbf{p}_j^T (\mathbf{T}_1 - \Sigma_0) \mathbf{p}_j + \delta_2 \mathbf{p}_j^T (\mathbf{T}_2 - \Sigma_0) \mathbf{p}_j]. \end{aligned}$$

Set $\frac{\partial R(\delta_1, \delta_2, \mathbf{D})}{\partial d_j} = 0$, then since we assumed $\delta_1 + \delta_2 \neq 1$,

$$\begin{aligned} d_j &= \frac{(1 - \delta_1 - \delta_2) \mathbf{p}_j^T \boldsymbol{\Sigma}_0 \mathbf{p}_j - \delta_1 \mathbf{p}_j^T (\mathbf{T}_1 - \boldsymbol{\Sigma}_0) \mathbf{p}_j - \delta_2 \mathbf{p}_j^T (\mathbf{T}_2 - \boldsymbol{\Sigma}_0) \mathbf{p}_j}{1 - \delta_1 - \delta_2} \\ &= \frac{1}{1 - \delta_1 - \delta_2} (\mathbf{p}_j^T \boldsymbol{\Sigma}_0 \mathbf{p}_j - \delta_1 \mathbf{p}_j^T \mathbf{T}_1 \mathbf{p}_j - \delta_2 \mathbf{p}_j^T \mathbf{T}_2 \mathbf{p}_j), \text{ so that} \\ \mathbf{D} &= \frac{1}{1 - \delta_1 - \delta_2} (\text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) - \delta_1 \text{diag}(\mathbf{P}^T \mathbf{T}_1 \mathbf{P}) - \delta_2 \text{diag}(\mathbf{P}^T \mathbf{T}_2 \mathbf{P})). \end{aligned} \quad (\text{S.11})$$

This proves the first part of Theorem 4. Equation (S.11) can be substituted into $R(\delta_1, \delta_2, \mathbf{D})$ in (S.10) for solving for the optimal δ_1 and δ_2 then. We have

$$\begin{aligned} R(\delta_1, \delta_2, \mathbf{D}) &= R(\delta_1, \delta_2) = \left\| \mathbf{P} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) \mathbf{P}^T - \delta_1 \widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_1} - \delta_2 \widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_2} + \delta_1 \mathbf{T}_1 + \delta_2 \mathbf{T}_2 - \boldsymbol{\Sigma}_0 \right\|_F^2 \\ &= \left\| (\mathbf{P} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) \mathbf{P}^T - \boldsymbol{\Sigma}_0) - \delta_1 (\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_1} - \mathbf{T}_1) - \delta_2 (\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_2} - \mathbf{T}_2) \right\|_F^2 \\ &= \text{tr}(\mathbf{P} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) \mathbf{P}^T - \boldsymbol{\Sigma}_0)^2 + \delta_1^2 \text{tr}(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_1} - \mathbf{T}_1)^2 + \delta_2^2 \text{tr}(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_2} - \mathbf{T}_2)^2 \\ &\quad - 2\delta_1 \text{tr}[(\mathbf{P} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) \mathbf{P}^T - \boldsymbol{\Sigma}_0)(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_1} - \mathbf{T}_1)] + 2\delta_1 \delta_2 \text{tr}[(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_1} - \mathbf{T}_1)(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_2} - \mathbf{T}_2)] \\ &\quad - 2\delta_2 \text{tr}[(\mathbf{P} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) \mathbf{P}^T - \boldsymbol{\Sigma}_0)(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_2} - \mathbf{T}_2)], \end{aligned} \quad (\text{S.12})$$

where $\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_i} = \mathbf{P} \text{diag}(\mathbf{P}^T \mathbf{T}_i \mathbf{P}) \mathbf{P}^T, i = 1, 2$. To find the optimal δ_1 and δ_2 , we take the partial derivative of R with respect to δ_1 and δ_2 respectively,

$$\begin{aligned} \frac{\partial R(\delta_1, \delta_2)}{\partial \delta_1} &= 2\delta_1 \text{tr}(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_1} - \mathbf{T}_1)^2 - 2\text{tr}[(\mathbf{P} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) \mathbf{P}^T - \boldsymbol{\Sigma}_0)(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_1} - \mathbf{T}_1)] \\ &\quad + 2\delta_2 \text{tr}(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_1} - \mathbf{T}_1)(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_2} - \mathbf{T}_2), \\ \frac{\partial R(\delta_1, \delta_2)}{\partial \delta_2} &= 2\delta_2 \text{tr}(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_2} - \mathbf{T}_2)^2 - 2\text{tr}[(\mathbf{P} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) \mathbf{P}^T - \boldsymbol{\Sigma}_0)(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_2} - \mathbf{T}_2)] \\ &\quad + 2\delta_1 \text{tr}(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_1} - \mathbf{T}_1)(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_2} - \mathbf{T}_2). \end{aligned}$$

Setting $\frac{\partial R(\delta_1, \delta_2)}{\partial \delta_1} = 0$ and $\frac{\partial R(\delta_1, \delta_2)}{\partial \delta_2} = 0$, we get

$$\begin{aligned} \delta_1 \text{tr}(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_1} - \mathbf{T}_1)^2 + \delta_2 \text{tr}[(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_1} - \mathbf{T}_1)(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_2} - \mathbf{T}_2)] &= \text{tr}[(\mathbf{P} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) \mathbf{P}^T - \boldsymbol{\Sigma}_0)(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_1} - \mathbf{T}_1)], \\ \delta_2 \text{tr}(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_2} - \mathbf{T}_2)^2 + \delta_1 \text{tr}[(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_1} - \mathbf{T}_1)(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_2} - \mathbf{T}_2)] &= \text{tr}[(\mathbf{P} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) \mathbf{P}^T - \boldsymbol{\Sigma}_0)(\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_2} - \mathbf{T}_2)]. \end{aligned}$$

Similar to (S.4), $\text{tr}[\mathbf{P} \text{diag}(\mathbf{P}^T \boldsymbol{\Sigma}_0 \mathbf{P}) \mathbf{P}^T (\widehat{\boldsymbol{\Sigma}}_{\mathbf{T}_i} - \mathbf{T}_i)] = 0, i = 1, 2$. Solving for δ_1 and δ_2 , we get the results as stated in Theorem 4. The denominator in δ_1 and δ_2 is not 0 since $\mathbf{T}_1, \mathbf{T}_2$ and $\mathbf{T}_1 - \mathbf{T}_2$ are not of the form $\mathbf{P} \mathbf{D} \mathbf{P}^T$ for some diagonal matrix $\mathbf{D}$. $\square$

Proof of Theorem 5. Let Assumptions (A1) and (A2) be satisfied and the data is not from a factor

model. Write $\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i} = \mathbf{Q}_i \mathbf{D}_{\mathbf{T}_i} \mathbf{Q}_i^T$, where $\mathbf{Q}_i = (\mathbf{q}_{i1}, \dots, \mathbf{q}_{ip})$ is orthogonal and $\mathbf{D}_{\mathbf{T}_i} = \text{diag}(d_{i1}, \dots, d_{ip})$.

Then for $i, j = 1, 2$, using the notation $a_{ij} = \text{tr}[(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})(\mathbf{T}_j - \tilde{\Sigma}_{\mathbf{T}_j})]$,

$$\begin{aligned} |\hat{\delta}_i - \delta_i| &\leq \frac{|\text{tr}[(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})(\Sigma_0 - \tilde{\Sigma}_2)]a_{3-i,3-i}| + |\text{tr}[(\mathbf{T}_{3-i} - \tilde{\Sigma}_{\mathbf{T}_{3-i}})(\Sigma_0 - \tilde{\Sigma}_2)]a_{12}|}{a_{22}a_{11} - a_{12}^2} \\ &\leq \frac{\left| \frac{\text{tr}[(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})(\Sigma_0 - \tilde{\Sigma}_2)]}{\text{tr}(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})^2} \cdot p^{-2}a_{11}a_{22} \right| + \left| \frac{\text{tr}[(\mathbf{T}_{3-i} - \tilde{\Sigma}_{\mathbf{T}_{3-i}})(\Sigma_0 - \tilde{\Sigma}_2)]}{\text{tr}(\mathbf{T}_{3-i} - \tilde{\Sigma}_{\mathbf{T}_{3-i}})^2} \cdot p^{-2}a_{12}a_{3-i,3-i} \right|}{p^{-2}(a_{22}a_{11} - a_{12}^2)} \rightarrow 0 \end{aligned}$$

in probability/almost surely, where we have used the same lines of proof in (S.5), and the assumption that $p^{-1}a_{ii}, p^{-2}(a_{22}a_{11} - a_{12}^2) \not\rightarrow 0$ and are finite. This proves $\hat{\delta}_i - \delta_i \rightarrow 0$ in probability/almost surely.

Now assume $\|\mathbf{T}_1 - \Sigma_0\| \rightarrow 0$ in probability/almost surely. Then using $\text{tr}[(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})\tilde{\Sigma}_{\mathbf{T}_j}] = 0$ for $i, j = 1, 2$,

$$\begin{aligned} \hat{\delta}_1 &= \frac{\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})(\Sigma_0 - \mathbf{T}_1)]a_{22} + a_{11}a_{22} - \text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})(\Sigma_0 - \mathbf{T}_1)]a_{12} - a_{12}^2}{a_{22}a_{11} - a_{12}^2}, \text{ with} \\ |\hat{\delta}_1 - 1| &\leq \frac{(p^{-1}a_{11})^{1/2} \cdot p^{-1}a_{22} \cdot \|\mathbf{T}_1 - \Sigma_0\| + (p^{-1}a_{22})^{1/2} \cdot |p^{-1}a_{12}| \cdot \|\mathbf{T}_1 - \Sigma_0\|}{p^{-2}(a_{22}a_{11} - a_{12}^2)} \rightarrow 0 \end{aligned}$$

in probability/almost surely, where we use $p^{-1}a_{ii}, p^{-2}(a_{22}a_{11} - a_{12}^2) \not\rightarrow 0$ and are finite. At the same time,

$$\begin{aligned} \hat{\delta}_2 &= \frac{\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})(\Sigma_0 - \mathbf{T}_1)]a_{11} + a_{12}a_{11} - \text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})(\Sigma_0 - \mathbf{T}_1)]a_{12} - a_{12}a_{11}}{a_{22}a_{11} - a_{12}^2}, \text{ with} \\ |\hat{\delta}_2| &\leq \frac{(p^{-1}a_{22})^{1/2} \cdot p^{-1}a_{11} \cdot \|\mathbf{T}_1 - \Sigma_0\| + (p^{-1}a_{11})^{1/2} \cdot |p^{-1}a_{12}| \cdot \|\mathbf{T}_1 - \Sigma_0\|}{p^{-2}(a_{22}a_{11} - a_{12}^2)} \rightarrow 0 \end{aligned}$$

in probability/almost surely. If $\|\mathbf{T}_2 - \Sigma_0\| \rightarrow 0$ in probability/almost surely, the lines to follow are exactly the same with the roles of $\hat{\delta}_1$ and $\hat{\delta}_2$ swapped. Hence we have proved that if $\|\mathbf{T}_i - \Sigma_0\| \rightarrow 0$ in probability/almost surely, we have $\hat{\delta}_i \rightarrow 1$ and $\hat{\delta}_{3-i} \rightarrow 0$ in probability/almost surely.

With $\|\mathbf{T}_i - \Sigma_0\| \rightarrow 0$ in probabiliy/almost surely assumed, consider

$$\begin{aligned} \|\Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2) - \Sigma_0\| &\leq \|\mathbf{P}_1 \text{diag}(\mathbf{P}_1^T \tilde{\Sigma}_2 \mathbf{P}_1) \mathbf{P}_1^T - \tilde{\Sigma}_{\mathbf{T}_i}\| + \|\mathbf{T}_i - \Sigma_0\| \\ &\quad + |\hat{\delta}_i - 1| \cdot \|\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i}\| + |\hat{\delta}_{3-i}| \cdot \|\mathbf{T}_{3-i} - \tilde{\Sigma}_{\mathbf{T}_{3-i}}\| \\ &\leq \max_{1 \leq i \leq p} \left| \frac{\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_0 \mathbf{p}_{1i}}{\mathbf{p}_{1i}^T \Sigma_0 \mathbf{p}_{1i}} \right| \cdot \|\Sigma_0\| + 2\|\mathbf{T}_i - \Sigma_0\| \\ &\quad + 2|\hat{\delta}_i - 1| \cdot \|\mathbf{T}_i\| + 2|\hat{\delta}_{3-i}| \cdot \|\mathbf{T}_{3-i}\| \rightarrow 0 \end{aligned}$$

in probability/almost surely, where we used Lemma 1 of Lam (2016) and Assumption (A2), as well as the finiteness of $\|\mathbf{T}_i\|$ in probability/almost surely.

For data from a factor model with Assumptions (F1) and (F2) in place, the proof follows exactly the same as before, except that we now have $p^{-2}a_{ii}, p^{-4}(a_{11}a_{22} - a_{12}^2) \not\rightarrow 0$ while $p^{-1}\|\Sigma_0\|, p^{-1}\|\mathbf{T}_i\|$ are finite in probability/almost surely. This completes the proof of the theorem. $\square$

Proof of Theorem 6. Recall the notations $a_{ij} = \text{tr}[(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})(\mathbf{T}_j - \tilde{\Sigma}_{\mathbf{T}_j})]$ and $b_{ij} = \text{tr}[(\mathbf{T}_i - \hat{\Sigma}_{\mathbf{T}_i})(\mathbf{T}_j - \hat{\Sigma}_{\mathbf{T}_j})]$ for $i, j = 1, 2$. In this proof, we define δ_i for $i = 1, 2$ which corresponds to the Σ_{Ideal} constructed using $\mathbf{P}$ rather than $\mathbf{P}_1$, and at the same time with knowledge of Σ_0 itself. Hence we have

$$\begin{aligned}\delta_i &= \frac{b_{3-i,3-i}\text{tr}[(\mathbf{T}_i - \hat{\Sigma}_{\mathbf{T}_i})\Sigma_0] - b_{12}\text{tr}[(\mathbf{T}_{3-i} - \hat{\Sigma}_{\mathbf{T}_{3-i}})\Sigma_0]}{b_{22}b_{11} - b_{12}^2}, \\ \hat{\delta}_i &= \frac{a_{3-i,3-i}\text{tr}[(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})\tilde{\Sigma}_2] - a_{12}\text{tr}[(\mathbf{T}_{3-i} - \tilde{\Sigma}_{\mathbf{T}_{3-i}})\tilde{\Sigma}_2]}{a_{22}a_{11} - a_{12}^2}.\end{aligned}$$

We first consider

$$\begin{aligned}\frac{1}{p}\|\Sigma_{\text{Ideal}} - \Sigma_0\|_F^2 &= \frac{1}{p}\|\mathbf{P}\text{diag}(\mathbf{P}^\top \Sigma_0 \mathbf{P})\mathbf{P}^\top - \Sigma_0 + \delta_1(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1}) + \delta_2(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\|_F^2 \\ &= \frac{1}{p}\|\text{diag}(\mathbf{P}^\top \Sigma_0 \mathbf{P}) - \mathbf{P}^\top \Sigma_0 \mathbf{P}\|_F^2 + \frac{1}{p}\|\delta_1(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1}) + \delta_2(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\|_F^2 \\ &\quad + \frac{2}{p}\text{tr}[(\mathbf{P}\text{diag}(\mathbf{P}^\top \Sigma_0 \mathbf{P})\mathbf{P}^\top - \Sigma_0)(\delta_1(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1}) + \delta_2(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2}))] \\ &= \frac{1}{p}\|\text{diag}(\mathbf{P}^\top \Sigma_0 \mathbf{P}) - \mathbf{P}^\top \Sigma_0 \mathbf{P}\|_F^2 + \frac{1}{p}(\delta_1^2 b_{11} + \delta_2^2 b_{22} + 2\delta_1 \delta_2 b_{12}) \\ &\quad - \frac{2}{p} \cdot (\delta_1 \text{tr}[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0] + \delta_2 \text{tr}[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\Sigma_0]) \\ &= \frac{1}{p}\left(\|\text{diag}(\mathbf{P}^\top \Sigma_0 \mathbf{P}) - \mathbf{P}^\top \Sigma_0 \mathbf{P}\|_F^2 - \delta_1 \text{tr}[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0] - \delta_2 \text{tr}[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\Sigma_0]\right),\end{aligned}$$

where the last equality uses the result

$$\delta_1^2 b_{11} + \delta_2^2 b_{22} + 2\delta_1 \delta_2 b_{12} = \delta_1 \text{tr}[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0] + \delta_2 \text{tr}[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\Sigma_0],$$

which can be proved with simple algebra.

Similarly, we have

$$\begin{aligned}
\frac{1}{p} \left\| \Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2 - \Sigma_0) \right\|_F^2 &= \frac{1}{p} \left\| \mathbf{P}_1 \text{diag}(\mathbf{P}_1^T \tilde{\Sigma}_2 \mathbf{P}_1) \mathbf{P}_1^T - \Sigma_0 + \hat{\delta}_1(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) + \hat{\delta}_2(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}) \right\|_F^2 \\
&= \frac{1}{p} \left\| \text{diag}(\mathbf{P}_1^T \tilde{\Sigma}_2 \mathbf{P}_1) - \mathbf{P}_1^T \Sigma_0 \mathbf{P}_1 \right\|_F^2 + \frac{1}{p} \left\| \hat{\delta}_1(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) + \hat{\delta}_2(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}) \right\|_F^2 \\
&\quad + \frac{2}{p} \text{tr}[(\mathbf{P}_1 \text{diag}(\mathbf{P}_1^T \tilde{\Sigma}_2 \mathbf{P}_1) \mathbf{P}_1^T - \Sigma_0)(\hat{\delta}_1(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) + \hat{\delta}_2(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}))] \\
&= \frac{1}{p} \sum_{r=1}^p (\mathbf{p}_{1r}^T \tilde{\Sigma}_2 \mathbf{p}_{1r} - \mathbf{p}_{1r}^T \Sigma_0 \mathbf{p}_{1r})^2 + \frac{1}{p} \left\| \text{diag}(\mathbf{P}_1^T \Sigma_0 \mathbf{P}_1) - \mathbf{P}_1^T \Sigma_0 \mathbf{P}_1 \right\|_F^2 \\
&\quad + \frac{1}{p} a_{22} \text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) \tilde{\Sigma}_2] \cdot \frac{\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) \tilde{\Sigma}_2] - 2 \text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) \Sigma_0]}{a_{22} a_{11} - a_{12}^2} \\
&\quad + \frac{1}{p} a_{11} \text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}) \tilde{\Sigma}_2] \cdot \frac{\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}) \tilde{\Sigma}_2] - 2 \text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}) \Sigma_0]}{a_{22} a_{11} - a_{12}^2} \\
&\quad + \frac{2}{p} a_{12} \left(\frac{\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}) \tilde{\Sigma}_2] \text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) \Sigma_0]}{a_{22} a_{11} - a_{12}^2} + \frac{\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) \tilde{\Sigma}_2] \text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}) \Sigma_0]}{a_{22} a_{11} - a_{12}^2} \right. \\
&\quad \left. - \frac{\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) \tilde{\Sigma}_2] \text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}) \tilde{\Sigma}_2]}{a_{22} a_{11} - a_{12}^2} \right) \\
&= R_1 + R_2 + R_3 + R_4, \quad \text{where} \\
R_1 &= \frac{1}{p} \left(\left\| \text{diag}(\mathbf{P}_1^T \Sigma_0 \mathbf{P}_1) - \mathbf{P}_1^T \Sigma_0 \mathbf{P}_1 \right\|_F^2 - \delta_1 \text{tr}[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1}) \Sigma_0] - \delta_2 \text{tr}[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2}) \Sigma_0] \right), \\
R_2 &= \frac{1}{p} \sum_{r=1}^p (\mathbf{p}_{1r}^T \tilde{\Sigma}_2 \mathbf{p}_{1r} - \mathbf{p}_{1r}^T \Sigma_0 \mathbf{p}_{1r})^2, \\
R_3 &= \frac{1}{p} \left((a_{22} \text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) \tilde{\Sigma}_2] - 2a_{12} \text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}) \tilde{\Sigma}_2]) \frac{\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) (\tilde{\Sigma}_2 - \Sigma_0)]}{a_{22} a_{11} - a_{12}^2} \right. \\
&\quad \left. + (a_{11} \text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}) \tilde{\Sigma}_2] - 2a_{12} \text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) \tilde{\Sigma}_2]) \frac{\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}) (\tilde{\Sigma}_2 - \Sigma_0)]}{a_{22} a_{11} - a_{12}^2} \right), \\
R_4 &= \frac{1}{p} \left(\frac{2a_{12} \text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) \tilde{\Sigma}_2] \text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}) \tilde{\Sigma}_2]}{a_{22} a_{11} - a_{12}^2} - \frac{a_{22} \text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) \tilde{\Sigma}_2] \text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1}) \Sigma_0]}{a_{22} a_{11} - a_{12}^2} \right. \\
&\quad \left. - \frac{a_{11} \text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}) \tilde{\Sigma}_2] \text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2}) \Sigma_0]}{a_{22} a_{11} - a_{12}^2} \right. \\
&\quad \left. + \delta_1 \text{tr}[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1}) \Sigma_0] + \delta_2 \text{tr}[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2}) \Sigma_0] \right).
\end{aligned}$$

With these, we can expand the efficiency loss as

$$EL(\Sigma_0, \Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2)) = 1 - \left(\frac{p^{-1} \left\| \Sigma(\mathbf{P}_1, \mathbf{T}, \tilde{\Sigma}_2) - \Sigma_0 \right\|_F^2}{p^{-1} \left\| \Sigma_{\text{Ideal}} - \Sigma_0 \right\|_F^2} \right)^{-1} = 1 - \left(\frac{R_1 + R_2 + R_3 + R_4}{p^{-1} \left\| \Sigma_{\text{Ideal}} - \Sigma_0 \right\|_F^2} \right)^{-1}.$$

Similar to the proof of Theorem 3, we have $\frac{R_1}{p^{-1} \left\| \Sigma_{\text{Ideal}} - \Sigma_0 \right\|_F^2} \rightarrow 1$ almost surely by Lemma S.4 of Lam (2016), and $R_2 \rightarrow 0$ almost surely by Lemma 1, following the exact lines of proof in (S.8). The proof completes if we can prove that $R_3, R_4 \rightarrow 0$ in probability/almost surely.

To do so, we prove several intermediate results first. Firstly,

$$\begin{aligned} \frac{1}{p}|a_{ij} - b_{ij}| &= \frac{1}{p}|\text{tr}[(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})(\mathbf{T}_j - \tilde{\Sigma}_{\mathbf{T}_j})] - \text{tr}[(\mathbf{T}_i - \hat{\Sigma}_{\mathbf{T}_i})(\mathbf{T}_j - \hat{\Sigma}_{\mathbf{T}_j})]| = \frac{1}{p}|\text{tr}[(\hat{\Sigma}_{\mathbf{T}_i} - \tilde{\Sigma}_{\mathbf{T}_i})\mathbf{T}_j]| \\ &\leq \frac{\|\mathbf{T}_i\|}{p} \sum_{r=1}^p |\mathbf{p}_r^\top \mathbf{T}_j \mathbf{p}_r - \mathbf{p}_{1r}^\top \mathbf{T}_j \mathbf{p}_{1r}| + \frac{\|\mathbf{T}_j\|}{p} \sum_{r=1}^p |\mathbf{p}_r^\top \mathbf{T}_i \mathbf{p}_r - \mathbf{p}_{1r}^\top \mathbf{T}_i \mathbf{p}_{1r}| \rightarrow 0 \end{aligned} \quad (\text{S.13})$$

almost surely for $i, j = 1, 2$ by Lemma 2, and the assumption that $\|\mathbf{T}_i\|$ is finite in probability/almost surely. Secondly,

$$\frac{1}{p}|\text{tr}[(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})(\Sigma_0 - \tilde{\Sigma}_2)]| = \frac{1}{p}a_{ii} \left| \frac{\text{tr}[(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})(\Sigma_0 - \tilde{\Sigma}_2)]}{\text{tr}(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})^2} \right| \rightarrow 0 \quad (\text{S.14})$$

in probability/almost surely for $i = 1, 2$ by (S.5), and the assumption that $p^{-1}a_{ii}$ is finite in probability/almost surely. Also, in probability/almost surely,

$$\frac{1}{p}|\text{tr}[(\mathbf{T}_i - \hat{\Sigma}_{\mathbf{T}_i})\Sigma_0]| \leq \left(\frac{1}{p}\text{tr}(\mathbf{T}_i - \hat{\Sigma}_{\mathbf{T}_i})^2 \right)^{1/2} \cdot \left(\frac{1}{p}\text{tr}(\Sigma_0^2) \right)^{1/2} \leq (p^{-1}b_{ii})^{1/2} \cdot \|\Sigma_0\| < \infty \quad (\text{S.15})$$

by the assumptions that $p^{-1}b_{ii}$ and $\|\Sigma_0\|$ are both finite. Similarly, we have $\frac{1}{p}|\text{tr}[(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})\Sigma_0]| < \infty$ in probability/almost surely. Combining these results with (S.14), we get

$$\frac{1}{p}|\text{tr}[(\mathbf{T}_i - \hat{\Sigma}_{\mathbf{T}_i})\tilde{\Sigma}_2]| \leq \frac{1}{p}|\text{tr}[(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})(\Sigma_0 - \tilde{\Sigma}_2)]| + \frac{1}{p}|\text{tr}[(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})\Sigma_0]| < \infty \quad (\text{S.16})$$

in probability/almost surely. At the same time,

$$\begin{aligned} p^{-2}|\text{tr}^2[(\mathbf{T}_i - \hat{\Sigma}_{\mathbf{T}_i})\Sigma_0] - \text{tr}^2[(\mathbf{T}_i - \tilde{\Sigma}_{\mathbf{T}_i})\Sigma_0]| &\leq |p^{-1}\text{tr}[(\hat{\Sigma}_{\mathbf{T}} - \tilde{\Sigma}_{\mathbf{T}})\Sigma_0] + 2p^{-1}\text{tr}[(\mathbf{T}_i - \hat{\Sigma}_{\mathbf{T}_i})\Sigma_0]| \\ &\quad \cdot p^{-1}|\text{tr}[(\hat{\Sigma}_{\mathbf{T}} - \tilde{\Sigma}_{\mathbf{T}})\Sigma_0]| \rightarrow 0 \end{aligned} \quad (\text{S.17})$$

in probability/almost surely by (S.9) and (S.15).

We can then bound R_3 by noting that

$$\begin{aligned} |R_3| &\leq \left(\frac{1}{p}a_{22} \cdot \frac{1}{p}|\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\tilde{\Sigma}_2]| + \frac{2}{p}|a_{12}| \cdot \frac{1}{p}|\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})\tilde{\Sigma}_2]| \right) \frac{p^{-1}|\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})(\tilde{\Sigma}_2 - \Sigma_0)]|}{p^{-2}|a_{22}a_{11} - a_{12}^2|} \\ &\quad + \left(\frac{1}{p}a_{11} \cdot \frac{1}{p}|\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})\tilde{\Sigma}_2]| + \frac{2}{p}|a_{12}| \cdot \frac{1}{p}|\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\tilde{\Sigma}_2]| \right) \frac{p^{-1}|\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})(\tilde{\Sigma}_2 - \Sigma_0)]|}{p^{-2}|a_{22}a_{11} - a_{12}^2|} \\ &\rightarrow 0 \end{aligned}$$

in probability/almost surely by the results in (S.14) and (S.16), along with the assumptions for the finiteness of $p^{-1}a_{ij}$ and $p^{-2}(a_{22}a_{11} - a_{12}^2)$ in probability/almost surely for $i, j = 1, 2$.

Also, we can decompose R_4 further. Consider

$$\begin{aligned}
R_4 &= \frac{1}{p} \left(\frac{2a_{12}\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\tilde{\Sigma}_2]\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})\tilde{\Sigma}_2]}{a_{22}a_{11} - a_{12}^2} - \frac{a_{22}\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\tilde{\Sigma}_2]\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\Sigma_0]}{a_{22}a_{11} - a_{12}^2} \right. \\
&\quad - \frac{a_{11}\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})\tilde{\Sigma}_2]\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})\Sigma_0]}{a_{22}a_{11} - a_{12}^2} - \frac{2b_{12}\text{tr}[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0]\text{tr}[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\Sigma_0]}{b_{22}b_{11} - b_{12}^2} \\
&\quad \left. + \frac{b_{22}\text{tr}^2[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0] + b_{11}\text{tr}^2[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\Sigma_0]}{b_{22}b_{11} - b_{12}^2} \right) \\
&= \frac{2p^{-3}a_{12}\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\tilde{\Sigma}_2]\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})\tilde{\Sigma}_2] - p^{-3}a_{22}\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\tilde{\Sigma}_2]\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\Sigma_0]}{p^{-2}(a_{22}a_{11} - a_{12}^2)} \\
&\quad + \frac{-p^{-3}a_{11}\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})\tilde{\Sigma}_2]\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})\Sigma_0] - 2p^{-3}b_{12}\text{tr}[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0]\text{tr}[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\Sigma_0]}{p^{-2}(a_{22}a_{11} - a_{12}^2)} \\
&\quad + \frac{p^{-3}(b_{22}\text{tr}^2[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0] + b_{11}\text{tr}^2[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\Sigma_0])}{p^{-2}(a_{22}a_{11} - a_{12}^2)} \\
&\quad - \frac{p^{-1}(\delta_1\text{tr}[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0] + \delta_2\text{tr}[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\Sigma_0]) \cdot p^{-2}(b_{22}b_{11} - b_{12}^2 - a_{22}a_{11} + a_{12}^2)}{p^{-2}(a_{22}a_{11} - a_{12}^2)} \\
&= \frac{R_{4,1} + R_{4,2} + 2R_{4,3}}{p^{-2}(a_{22}a_{11} - a_{12}^2)} + \frac{R_{4,4} \cdot R_{4,5}}{p^{-2}(a_{22}a_{11} - a_{12}^2)}, \quad \text{where} \\
R_{4,1} &= p^{-3}b_{22}\text{tr}^2[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0] - p^{-3}a_{22}\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\tilde{\Sigma}_2]\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\Sigma_0], \\
R_{4,2} &= p^{-3}b_{11}\text{tr}^2[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\Sigma_0] - p^{-3}a_{11}\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})\tilde{\Sigma}_2]\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})\Sigma_0], \\
R_{4,3} &= p^{-3}a_{12}\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\tilde{\Sigma}_2]\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})\tilde{\Sigma}_2] - p^{-3}b_{12}\text{tr}[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0]\text{tr}[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\Sigma_0], \\
R_{4,4} &= p^{-2}(b_{22}b_{11} - b_{12}^2 - a_{22}a_{11} + a_{12}^2), \\
R_{4,5} &= -p^{-1}(\delta_1\text{tr}[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0] + \delta_2\text{tr}[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\Sigma_0]).
\end{aligned}$$

To bound $R_{4,1}$, consider

$$\begin{aligned}
|R_{4,1}| &= |p^{-3}(b_{22} - a_{22})\text{tr}^2[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0] + p^{-3}a_{22}(\text{tr}^2[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0] - \text{tr}^2[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\Sigma_0]) \\
&\quad + p^{-3}a_{22}\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\Sigma_0]\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})(\Sigma_0 - \tilde{\Sigma}_2)]| \\
&\leq p^{-1}|b_{22} - a_{22}| \cdot p^{-2}\text{tr}^2[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0] + p^{-1}a_{22} \cdot p^{-2}|\text{tr}^2[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0] - \text{tr}^2[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\Sigma_0]| \\
&\quad + p^{-1}a_{22} \cdot p^{-1}|\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\Sigma_0]| \cdot p^{-1}|\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})(\Sigma_0 - \tilde{\Sigma}_2)]| \rightarrow 0
\end{aligned}$$

in probability/almost surely by the results of (S.13), (S.14), (S.15), (S.16) and (S.17), together with the assumption that $p^{-1}a_{22}$ is finite. Follow the exactly the same proof, we can easily get $R_{4,2} \rightarrow 0$ in

probability/almost surely. Furthermore,

$$\begin{aligned}
|R_{4,3}| &\leq p^{-1}|b_{12}| \cdot p^{-1}|\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})\tilde{\Sigma}_2]| \cdot (p^{-1}|\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})(\tilde{\Sigma}_2 - \Sigma_0)]| + p^{-1}|\text{tr}[(\hat{\Sigma}_{\mathbf{T}_1} - \tilde{\Sigma}_{\mathbf{T}_1})\Sigma_0]|) \\
&\quad + p^{-1}|b_{12}| \cdot p^{-1}|\text{tr}[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0]| \cdot (p^{-1}|\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})(\tilde{\Sigma}_2 - \Sigma_0)]| + p^{-1}|\text{tr}[(\hat{\Sigma}_{\mathbf{T}_2} - \tilde{\Sigma}_{\mathbf{T}_2})\Sigma_0]|) \\
&\quad + p^{-1}|a_{12} - b_{12}| \cdot p^{-1}|\text{tr}[(\mathbf{T}_1 - \tilde{\Sigma}_{\mathbf{T}_1})\tilde{\Sigma}_2]| \cdot p^{-1}|\text{tr}[(\mathbf{T}_2 - \tilde{\Sigma}_{\mathbf{T}_2})\tilde{\Sigma}_2]| \rightarrow 0
\end{aligned}$$

in probability/almost surely by the results of (S.9), (S.13), (S.14), (S.15) and (S.16), together with the assumption that $p^{-1}b_{12}$ is finite in probability/almost surely.

$$\begin{aligned}
|R_{4,4}| &= p^{-2}(b_{22}b_{11} - b_{12}^2 - a_{22}a_{11} + a_{12}^2) = p^{-2}(b_{11}(b_{22} - a_{22}) + a_{22}(b_{11} - a_{11}) + (a_{12} + b_{12})(a_{12} - b_{12})) \\
&\leq p^{-1}b_{11} \cdot p^{-1}|b_{22} - a_{22}| + p^{-1}a_{22} \cdot p^{-1}|b_{11} - a_{11}| + p^{-1}|b_{12} + a_{12}| \cdot p^{-1}|b_{12} - a_{12}| \rightarrow 0
\end{aligned}$$

almost surely by the result of (S.13) and the assumption that $p^{-1}a_{ij}, p^{-1}b_{ij}$ are all finite in probability/almost surely. Also, in probability/almost surely,

$$\begin{aligned}
R_{4,5} &= \frac{2p^{-1}b_{12} \cdot p^{-1}\text{tr}[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0] \cdot p^{-1}\text{tr}[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\Sigma_0]}{p^{-2}(b_{22}b_{11} - b_{12}^2)} \\
&\quad - \frac{p^{-1}b_{22} \cdot p^{-2}\text{tr}^2[(\mathbf{T}_1 - \hat{\Sigma}_{\mathbf{T}_1})\Sigma_0] + p^{-1}b_{11} \cdot p^{-2}\text{tr}^2[(\mathbf{T}_2 - \hat{\Sigma}_{\mathbf{T}_2})\Sigma_0]}{p^{-2}(b_{22}b_{11} - b_{12}^2)} < \infty
\end{aligned}$$

by the result of (S.15) and the assumption that $p^{-1}b_{ij}$ and $p^{-2}(b_{22}b_{11} - b_{12}^2)$ are all finite in probability/almost surely for $i, j = 1, 2$.

So, $R_{4,5}$ is finite. By the assumption for the finiteness of $p^{-2}(a_{22}a_{11} - a_{12}^2)$ in probability/almost surely, we can finally conclude that $R_4 \rightarrow 0$ in probability/almost surely. This completes the proof of the theorem.

□

Proof of Theorem 7. For Frobenius loss,

$$\left\| \hat{\Sigma}_{m,M} - \Sigma_0 \right\|_F^2 = \left\| \frac{1}{M} \sum_{j=1}^M (\hat{\Sigma}_m^{(j)} - \Sigma_0) \right\|_F^2 \leq \left(\frac{1}{M} \sum_{j=1}^M \left\| \hat{\Sigma}_m^{(j)} - \Sigma_0 \right\|_F \right)^2 \leq \frac{1}{M} \sum_{j=1}^M \left\| \hat{\Sigma}_m^{(j)} - \Sigma_0 \right\|_F^2, \quad (\text{S.18})$$

so that,

$$EL(\Sigma_0, \hat{\Sigma}_{m,M}) \leq 1 - \frac{\left\| \hat{\Sigma}_{\text{Ideal}} - \Sigma_0 \right\|_F^2}{\frac{1}{M} \sum_{j=1}^M \left\| \hat{\Sigma}_m^{(j)} - \Sigma_0 \right\|_F^2} \leq 1 - \frac{1}{\frac{1}{M} \sum_{j=1}^M \frac{1}{1 - EL(\Sigma_0, \hat{\Sigma}_m^{(j)})}} \rightarrow 0. \quad (\text{S.19})$$

The last step follows Theorem 6 that, if $p^{-1}a_{11}, p^{-1}a_{22}$ and $p^{-2}(a_{11}a_{22} - a_{12}^2) \not\rightarrow 0$ in probability/almost surely, then $EL(\Sigma_0, \hat{\Sigma}_m^{(j)}) \rightarrow 0$ in probability/almost surely. This proves the first part of the theorem.

To prove the remaining part of theorem, for $i = 1, 2$ and $j = 1, \dots, M$, we note first that

$$\text{tr}(\tilde{\mathbf{\Sigma}}_{\mathbf{T}_{ij}}) = \text{tr}(\mathbf{P}_1^T \text{diag}(\mathbf{P}_{1j} \mathbf{T}_{ij} \mathbf{P}_{1j}^T) \mathbf{P}_{1j}) = \text{tr}(\text{diag}(\mathbf{P}_{1j} \mathbf{T}_{ij} \mathbf{P}_{1j}^T)) = \text{tr}(\mathbf{P}_{1j} \mathbf{T}_{ij} \mathbf{P}_{1j}^T) = \text{tr}(\mathbf{T}_{ij}). \quad (\text{S.20})$$

Using (S.20), with $\hat{\delta}_{ij}$ being finite in probability/almost surely since the corresponding $p^{-1}a_{11}, p^{-1}a_{22}$ and $p^{-2}(a_{11}a_{22} - a_{12}^2) \not\rightarrow 0$ in probability/almost surely,

$$\begin{aligned} \frac{1}{p}[\text{tr}(\hat{\mathbf{\Sigma}}_{m,M}) - \text{tr}(\mathbf{\Sigma}_0)] &= \frac{1}{p} \left[\text{tr} \left(\frac{1}{M} \sum_{j=1}^M \mathbf{P}_{1j} \text{diag}(\mathbf{P}_{1j}^T \tilde{\mathbf{\Sigma}}_2^{(j)} \mathbf{P}_{1j}) \mathbf{P}_{1j}^T \right) - \text{tr}(\mathbf{\Sigma}_0) \right] \\ &= \frac{1}{pM} \sum_{j=1}^M [\text{tr}(\mathbf{P}_{1j} \text{diag}(\mathbf{P}_{1j}^T \tilde{\mathbf{\Sigma}}_2^{(j)} \mathbf{P}_{1j}) \mathbf{P}_{1j}^T) - \text{tr}(\mathbf{\Sigma}_0)] \\ &= \frac{1}{M} \sum_{j=1}^M \frac{1}{p} [\text{tr}(\mathbf{P}_{1j}^T \tilde{\mathbf{\Sigma}}_2^{(j)} \mathbf{P}_{1j}) - \text{tr}(\mathbf{P}_{1j}^T \mathbf{\Sigma}_0 \mathbf{P}_{1j})] \rightarrow 0, \end{aligned} \quad (\text{S.21})$$

where the last convergence step follows from Theorem 1 of Lam (2016) if Assumptions (A1), (A2) are satisfied, and it follows from Theorem 3 of Lam (2016) if the data follows a factor model with Assumptions (F1), (F2) satisfied.

Finally, to prove the last inequality, we first consider data under Assumptions (A1) and (A2) first. Observe that

$$\begin{aligned} \frac{1}{p} \text{tr}(\hat{\mathbf{\Sigma}}_{m,M} \mathbf{\Sigma}_0) &= \frac{1}{M} \sum_{j=1}^M \frac{1}{p} \left(\text{tr}(\mathbf{P}_{1j} \text{diag}(\mathbf{P}_{1j}^T \tilde{\mathbf{\Sigma}}_2^{(j)} \mathbf{P}_{1j}) \mathbf{P}_{1j}^T \mathbf{\Sigma}_0) + \hat{\delta}_{1j} \text{tr}[(\mathbf{T}_{1j} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}_{1j}}) \mathbf{\Sigma}_0] \right. \\ &\quad \left. + \hat{\delta}_{2j} \text{tr}[(\mathbf{T}_{2j} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}_{2j}}) \mathbf{\Sigma}_0] \right) = \frac{1}{M} \sum_{j=1}^M \sum_{i=1}^5 R_{ij}, \quad \text{where} \\ R_{1j} &= \frac{1}{p} \text{tr}(\mathbf{P}_{1j} \text{diag}(\mathbf{P}_{1j}^T \tilde{\mathbf{\Sigma}}_2^{(j)} \mathbf{P}_{1j}) \mathbf{P}_{1j}^T \mathbf{\Sigma}_0), \\ R_{2j} &= \frac{1}{p} \delta_{1j} \text{tr}[(\mathbf{T}_{1j} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}_{1j}}) \mathbf{\Sigma}_0], \quad R_{3j} = \frac{1}{p} \delta_{2j} \text{tr}[(\mathbf{T}_{2j} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}_{2j}}) \mathbf{\Sigma}_0], \\ R_{4j} &= \frac{1}{p} (\hat{\delta}_{1j} - \delta_{1j}) \text{tr}[(\mathbf{T}_{1j} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}_{1j}}) \mathbf{\Sigma}_0], \quad R_{5j} = \frac{1}{p} (\hat{\delta}_{2j} - \delta_{2j}) \text{tr}[(\mathbf{T}_{2j} - \tilde{\mathbf{\Sigma}}_{\mathbf{T}_{2j}}) \mathbf{\Sigma}_0]. \end{aligned}$$

We analyze each term above. Firstly, we have for $\mathbf{P}_{1j} = (\mathbf{p}_{1j,1}, \dots, \mathbf{p}_{1j,p})$,

$$\begin{aligned} R_{1j} &= R_{1j,1} + R_{1j,2}, \quad \text{where} \\ R_{1j,1} &= \frac{1}{p} \sum_{r=1}^p \left(\frac{\mathbf{p}_{1j,r}^T \tilde{\mathbf{\Sigma}}_2^{(j)} \mathbf{p}_{1j,r}}{\mathbf{p}_{1j,r}^T \mathbf{\Sigma}_0 \mathbf{p}_{1j,r}} - 1 \right) (\mathbf{p}_{1j,r}^T \mathbf{\Sigma}_0 \mathbf{p}_{1j,r})^2, \quad R_{1j,2} = \frac{1}{p} \sum_{r=1}^p (\mathbf{p}_{1j,r}^T \mathbf{\Sigma}_0 \mathbf{p}_{1j,r})^2. \end{aligned}$$

The first term has

$$|R_{1j,1}| \leq \max_{1 \leq r \leq p} \left| \frac{\mathbf{p}_{1j,r}^\top \tilde{\Sigma}_2^{(j)} \mathbf{p}_{1j,r}}{\mathbf{p}_{1j,r}^\top \Sigma_0 \mathbf{p}_{1j,r}} - 1 \right| \cdot \frac{1}{p} \sum_{r=1}^p (\mathbf{p}_{1j,r}^\top \Sigma_0 \mathbf{p}_{1j,r})^2 \leq \max_{1 \leq r \leq p} \left| \frac{\mathbf{p}_{1j,r}^\top \tilde{\Sigma}_2^{(j)} \mathbf{p}_{1j,r}}{\mathbf{p}_{1j,r}^\top \Sigma_0 \mathbf{p}_{1j,r}} - 1 \right| \cdot \lambda_{\max}^2(\Sigma_0) \rightarrow 0,$$

where the last step used the almost sure convergence result in Lemma 1 of Lam (2016), and that $\|\Sigma_0\| = O(1)$. We also have $R_{1j,2} \geq \lambda_{\min}^2(\Sigma_0)$, so that we have proved almost surely,

$$R_{1j} \geq \lambda_{\min}^2(\Sigma_0). \quad (\text{S.22})$$

Next, consider

$$\begin{aligned} p(R_{2j} + R_{3j}) &= \frac{\text{tr}^2[(\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}})\Sigma_0] \text{tr}(\mathbf{T}_{2j} - \tilde{\Sigma}_{\mathbf{T}_{2j}})^2 + \text{tr}^2[(\mathbf{T}_{2j} - \tilde{\Sigma}_{\mathbf{T}_{2j}})\Sigma_0] \text{tr}(\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}})^2}{\text{tr}(\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}})^2 \text{tr}(\mathbf{T}_{2j} - \tilde{\Sigma}_{\mathbf{T}_{2j}})^2 - \text{tr}^2[(\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}})(\mathbf{T}_{2j} - \tilde{\Sigma}_{\mathbf{T}_{2j}})]} \\ &\quad - \frac{2\text{tr}[(\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}})\Sigma_0] \text{tr}[(\mathbf{T}_{2j} - \tilde{\Sigma}_{\mathbf{T}_{2j}})\Sigma_0] \text{tr}[(\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}})(\mathbf{T}_{2j} - \tilde{\Sigma}_{\mathbf{T}_{2j}})]}{\text{tr}(\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}})^2 \text{tr}(\mathbf{T}_{2j} - \tilde{\Sigma}_{\mathbf{T}_{2j}})^2 - \text{tr}^2[(\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}})(\mathbf{T}_{2j} - \tilde{\Sigma}_{\mathbf{T}_{2j}})]} \\ &= \frac{\text{tr}\left((\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}}) \text{tr}[(\mathbf{T}_{2j} - \tilde{\Sigma}_{\mathbf{T}_{2j}})\Sigma_0] - (\mathbf{T}_{2j} - \tilde{\Sigma}_{\mathbf{T}_{2j}}) \text{tr}[(\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}})\Sigma_0]\right)^2}{\text{tr}(\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}})^2 \text{tr}(\mathbf{T}_{2j} - \tilde{\Sigma}_{\mathbf{T}_{2j}})^2 - \text{tr}^2[(\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}})(\mathbf{T}_{2j} - \tilde{\Sigma}_{\mathbf{T}_{2j}})]} \geq 0 \end{aligned}$$

in probability/almost surely, since the denominator above is also non-negative by the Cauchy-Schwarz inequality on the inner product $\text{tr}(\mathbf{AB})$ where $\mathbf{A}, \mathbf{B}$ are real square matrices, and by our assumptions it is positive in probability/almost surely.

Finally,

$$|R_{4j}| \leq |\hat{\delta}_{1j} - \delta_{1j}| \left(\frac{1}{p} \text{tr}(\mathbf{T}_{1j} - \tilde{\Sigma}_{\mathbf{T}_{1j}})^2 \right)^{1/2} \left(\frac{1}{p} \text{tr}(\Sigma_0^2) \right)^{1/2} \rightarrow 0$$

in probability/almost surely, which is the result of Theorem 5 and our assumptions that $p^{-1}a_{11} < \infty$ and $\|\Sigma_0\| = O(1)$. Similar result holds for $|R_{5j}|$. Hence combining (S.22) and the results we proved for R_{2j} to R_{5j} , the result follows.

It remains to show the trace property for data from a factor model under Assumptions (F1) and (F2). But in fact all the above steps follow, except we need to note that now $p^{-2}a_{ii}$ and $p^{-2}\lambda_{\max}(\Sigma_0)$ are finite. This completes the proof of the theorem. $\square$

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